

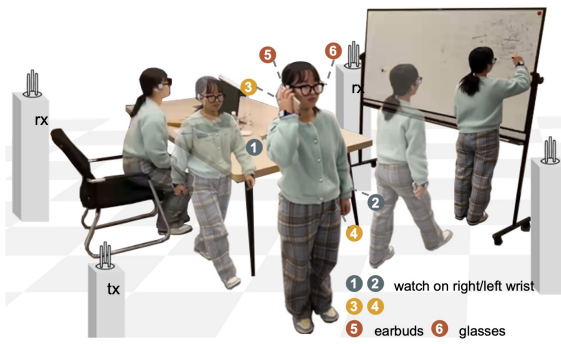
XRFv2 Plus: A Multimodal Sensor-Vision-Language Dataset for Action Understanding

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<https://github.com/airslab2020/XRFV2>



Synchronized release content. XRFv2 Plus provides synchronized sensing streams, pseudo body information, and action-understanding labels for the same video-aligned sequence set:

Modality	RGB embeddings; five-position IMU; AirPods IMU; 3 Wi-Fi CSI receivers; Depth maps; Infrared maps.
Pseudo body information	Human 2D Pose; Human 3D Pose; Human Mesh; DensePose.
Action understanding labels	Temporal action localization; Action captioning; Action QA.

Figure 1. Overview of XRFv2 Plus. The left panel illustrates the source sensing setup, while the right panel summarizes the synchronized modalities and sensor-vision-language annotation tasks in the released benchmark.

Abstract

public release organization.

We present XRFv2 Plus, a synchronized multimodal dataset for sensor-vision-language action understanding. Built from the XRFv2 recording corpus, XRFv2 Plus reorganizes 853 valid continuous action sequences around a common cropped-video timeline and releases aligned WiFi CSI, five-position IMU, AirPods IMU, RGB video embeddings, Kinect depth videos, Kinect infrared videos, 2D pose, depth-assisted 3D pose, SMPL mesh, and DensePose-style human-surface information. The dataset further provides relative-time temporal action localization annotations, action captioning annotations, and action question answering annotations. This paper does not introduce a new recording campaign; instead, it defines a new public benchmark built on a different release contract, modality set, annotation set, and task scope. XRFv2 Plus defines a unified video-aligned benchmark contract: standardized tensor shapes, fixed device order, per-second sensor resampling, privacy-aware no-RGB public packaging, and explicit handling of shortened Kinect-video cases. This paper describes the dataset construction, alignment protocol, modality formats, annotation schemas, and

1. XRFv2 Plus at a Glance

XRFv2 Plus is a synchronized multimodal action-understanding benchmark built from the XRFv2 recording corpus. All released signals and annotations share the same 853 video-aligned sequence IDs and use relative time as the common temporal axis. The benchmark synchronizes RGB embeddings, five-position IMU, AirPods IMU, 3 Wi-Fi CSI receivers, depth maps, infrared maps, Human 2D Pose, Human 3D Pose, Human Mesh, DensePose, temporal action localization, Action captioning, and Action QA. Figure 1 summarizes the release content and its relationship to the source sensing setup.

2. Relation to XRFv2 and Scope

The original XRFv2 paper introduced a multimodal indoor action-understanding corpus for temporal action localization (TAL) and action summarization using WiFi and wearable IMU signals, as described in arXiv:2501.01903 [2]. XRFv2 Plus is derived from the same recording campaign, but it defines a different public benchmark. Its goal is not to add

Table 1. Key differences between XRFv2 and XRFv2 Plus. XRFv2 Plus is derived from the original recording corpus but defines a new synchronized sensor-vision-language benchmark and public release contract.

Aspect	XRFv2	XRFv2 Plus
Primary focus	Multimodal action understanding with WiFi and wearable IMU signals, including TAL and action summarization.	Sensor-vision-language action understanding with synchronized sensing streams, visual body information, TAL, captioning, and QA.
Sample contract	Original candidate sequences and task-specific processing.	853 standardized video-aligned sequences with shared sample IDs and relative-time labels.
Time reference	Command-time and modality-specific processing are used in the original release.	Cropped Kinect RGB timeline is used as the reference; depth, IR, WiFi, IMU, AirPods, labels, and pseudo body information are aligned to it.
Released modalities	WiFi and IMU-centered release for the original tasks.	WiFi, five-position IMU, AirPods IMU, RGB embeddings, depth videos, IR videos, 2D pose, 3D pose, SMPL mesh, and DensePose-style information.
Language tasks	Action summarization-oriented annotations.	Action captioning annotations and 108,929 Action QA pairs across 21 question types.
Tensor convention	Not organized as a unified video-aligned tensor release.	Fixed tensor contracts: WiFi [50T, 3, 3, 30], IMU [50T, 5, 6], AirPods [25T, 6], RGB feature [T, 1024].
Public packaging	Original dataset release and task resources.	No-RGB public package with H5 files, annotations, depth/IR videos, and body-information files; raw Kinect and RGB videos are withheld for privacy.

a new collection session; instead, it converts the recordings into a stricter video-aligned, sensor-vision-language release in which all modalities and annotations share 853 sample IDs, cropped-video reference windows, and a relative-time convention.

The scope of this paper is the XRFv2 Plus benchmark contract: sample selection, synchronization rules, tensor shapes, sampling-rate conversion, shortened-video handling, annotation schemas, privacy-aware public packaging, and file organization. Each released sample is rebuilt around the cropped-video timeline and paired with WiFi CSI, five-position IMU, AirPods IMU, RGB video embeddings, cropped Kinect depth/infrared videos, body information, TAL segments, action captioning, and action QA. Table 1 summarizes the practical differences between XRFv2 and XRFv2 Plus.

3. Source Recordings and Command-Time

The source XRFv2 recordings were collected with 16 volunteers in three household scenes: dining room, study room, and bedroom [2]. Volunteers followed preset text-to-audio action prompts. The system recorded the issue and end time of each prompt; this prompt schedule is referred to as *command-time* and serves as the initial action-boundary reference before video alignment.

The sensing setup combines one WiFi transmitter and three receivers with Intel 5300 network cards, a five-position IMU setup, AirPods Pro, and Azure Kinect video. The five IMUs were mounted on the left wrist, right wrist, a phone case in the left pants pocket, a phone case in the right pants pocket used as the phone surrogate during phone-related

Table 2. Summary of the XRFv2 Plus release.

Item	Value
Standardized samples	853
Source candidate sequences	880
Excluded candidates	27 with incomplete five-position IMU coverage
Volunteers	16
Scenes	Diningroom, Studyroom, bedroom
Action classes	34 in the video-aligned annotations
Action segments	8157
Duration range	43–89 s per sequence
Actions per sample	7–11
Total duration in video-aligned release	16 h 14 min 38 s

actions, and one temple arm of a pair of glasses. WiFi packets were sent at 200 packets per second on channel 128 (5.64 GHz), with CSI extracted from 30 OFDM subcarriers. AirPods Pro data were recorded at approximately 25 Hz, the other IMU modules at 50 Hz, and Kinect streams at 15 FPS.

The source pool contains 880 candidate sequences. XRFv2 Plus standardizes 853 valid sequences; the remaining 27 candidates are excluded because they lack complete five-position IMU coverage. Table 2 summarizes the resulting release.

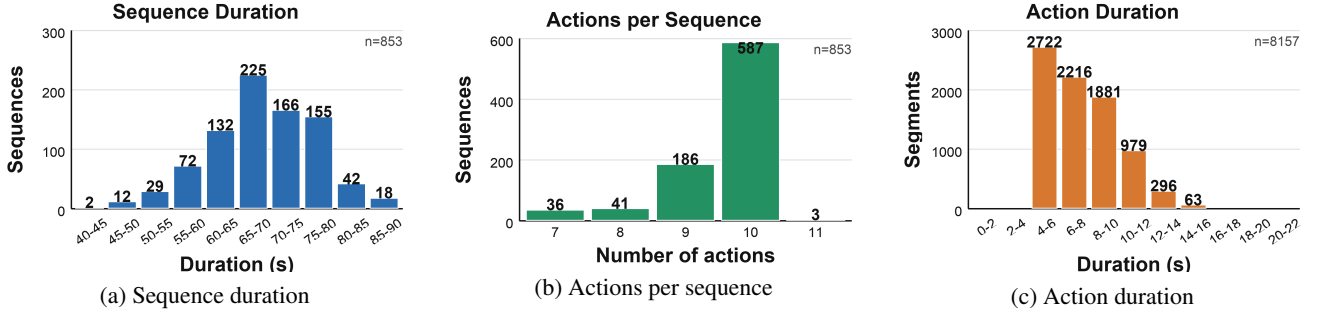


Figure 2. Temporal statistics of the 853 video-aligned sequences. Sequence duration is measured from the first action start to the last action end. Action duration is computed as the end time minus the start time of each TAL segment.

Table 3. Samples whose video-aligned annotations removed the last one or two actions.

Sample	Removed	Kept	RGB duration
6_bedroom_1	1	9/10	53.4 s
6_bedroom_11	1	9/10	60.933 s
6_bedroom_14	2	8/10	57.933 s
6_bedroom_2	2	8/10	58.933 s
6_bedroom_3	2	8/10	60.000 s
6_bedroom_4	2	8/10	61.000 s
6_bedroom_5	2	8/10	65.067 s

4. Video-Aligned Multimodal Data

All synchronized modalities are aligned to the same sample order through shared sample IDs and relative-time windows. In this release, one sample corresponds to one complete action sequence. The current video-aligned release was obtained by using the corrected cropped Kinect RGB videos and second-level RGB features as the reference timeline. Most cropped Kinect RGB videos cover the relative command-time window directly. For five samples with small video/command mismatches below roughly 1.3 seconds, the second-level RGB feature sequence was padded at the tail by repeating the last available feature vector. A separate set of seven bedroom samples had insufficient RGB video coverage for the complete command sequence. For these samples, the last one or two actions were removed from the video-aligned command-time annotation, as listed in Table 3. The corrected RGB videos and RGB feature sequences define the reference timeline for the release: depth videos, infrared videos, WiFi, five-position IMU, AirPods, TAL, captioning, and QA annotations are all segmented, trimmed, or resampled to the same cropped-video-aligned windows.

Figure 2 summarizes the temporal statistics of the resulting 853 video-aligned sequences, including full sequence duration, the number of action segments per sequence, and per-action segment duration. Table 4 then lists the main data files and tensor conventions used by the standardized release. This section first describes Kinect RGB video features, depth

videos, and infrared videos, followed by WiFi and wearable time-series streams. Figure 3 shows a synchronized visual example.

4.1. RGB Video Features

The RGB feature H5 stores one 1024-dimensional feature vector per second. Features were extracted from cropped Kinect RGB videos using TorchVision `torchvision.models.video.swin3d_b`, with `Swin3D_B_Weights.KINETICS400_IMAGENET22K_V1`. The classifier head was replaced with `torch.nn.Identity()`, and the default pre-processing transform from the TorchVision weight object was used. For each second, 15 RGB frames at 15 FPS were sampled as one clip and encoded by Swin3D-B [3]. The feature H5 is:

`data/video_feature_853_video_aligned.h5`.

4.2. Depth Videos

The cropped Kinect depth videos are kept at 15 FPS with 512×512 resolution and aligned to the same sequence IDs and cropped-video-aligned windows as the RGB feature H5 files and annotations. Each sample directory contains `depth_cropped.mkv` under `videos/kinect_rgb_ir_depth/<scene>/<sample_id>/`. For the 853 aligned samples, the expected archive contains 853 depth files with an estimated size of 57.63 GiB.

4.3. Infrared Videos

The cropped Kinect infrared videos are kept at 15 FPS with 512×512 resolution and released with the same directory structure and sample IDs as the RGB and depth streams. Each sample directory contains `ir_cropped.mkv`. For the 853 aligned samples, the expected archive contains 853 IR files with an estimated size of 94.02 GiB.

4.4. WiFi

The raw CSI recordings are temporally aligned to the action sequence and converted to a 50 Hz release. Each sample

Table 4. Main data files and tensor conventions in the standardized release.

Modality	Sampling / stride	Shape per sample	File
RGB video feature	1 vector/s	$[T, 1024]$	video_feature_853_video_aligned.h5
Kinect depth videos	15 FPS, 512×512	depth_cropped.mkv	videos/kinect_rgb_ir_depth/<scene>/<sample>/
Kinect infrared videos	15 FPS, 512×512	ir_cropped.mkv	videos/kinect_rgb_ir_depth/<scene>/<sample>/
WiFi CSI amplitude/phase	50 Hz	$[50T, 3, 3, 30]$	wifi_50hz_853_video_aligned.h5
Five-position IMU	50 Hz	$[50T, 5, 6]$	imu_50hz_853_video_aligned.h5
AirPods Pro IMU	25 Hz	$[25T, 6]$	airpods_25hz_853_video_aligned.h5

Figure 3. Synchronized visual example from sample 0_Diningroom_1. The row shows RGB, infrared, depth, 2D pose over RGB, depth-assisted 3D pose, SMPL/ROMP mesh over RGB, and a DensePose-style surface-color rendering over RGB, all generated from the same video frame. The face region in RGB-based panels is blurred for privacy.

stores amplitude and phase datasets, `amp` and `pha`, with shape $[50T, 3, 3, 30]$, corresponding to time, receiver, receiver antenna, and OFDM subcarrier dimensions. The release H5 is:

```
data/wifi_50hz_853_video_aligned.h5.
```

4.5. Five-Position IMU

The standardized IMU H5 stores five positions in a fixed order:

left wrist, right wrist, left pants-pocket phone case,
right pants-pocket phone case, glasses temple.

The two phone-case entries denote IMUs attached to phone cases placed in the left and right pants pockets, respectively; the right-side phone case also served as the simulated phone during phone-related actions. The glasses-temple device was attached to one temple arm of a pair of glasses. Each IMU entry has six channels: three-axis acceleration and three-axis gyroscope. The resulting tensor shape is $[50T, 5, 6]$. During preprocessing, raw IMU text files were first split by physical device ID and then segmented by the first action start time and the last action end time in the corresponding command-time JSON. The segmented signals were resampled per second to 50 Hz. Per-second resampling is important: each one-second interval is independently interpolated to exactly 50 samples, so integer second boundaries remain synchronized with TAL labels and video features. The release H5 is:

```
data/imu_50hz_853_video_aligned.h5.
```

Two physical IMU hardware sets were mapped to the same five semantic mounting positions. The first set maps 6770 to left wrist, 6649 to right wrist, 6748 to the phone case in the left pants pocket, 6678 to the phone case in the right pants pocket, and 6735 to the glasses temple. The second set maps 6751 to left wrist, 6676 to right wrist, 5126 to the phone case in the left pants pocket, 6633 to the phone case in the right pants pocket, and 6727 to the glasses temple.

4.6. AirPods

AirPods Pro recordings are represented with three-axis acceleration and three-axis gyroscope channels. Each segmented sequence is resampled per second to 25 Hz, producing a tensor of shape $[25T, 6]$. The release H5 is:

```
data/airpods_25hz_853_video_aligned.h5.
```

5. 2D Pose, 3D Pose, Mesh, and DensePose

The pose, mesh, and DensePose files are released as structured human-body information. They can be used either as pseudo-annotations for body-state analysis or as additional input modalities for multimodal action understanding, especially when models benefit from compact human-centric representations rather than raw RGB frames.

5.1. 2D Pose

Per-frame 2D pose information is extracted from cropped RGB videos using `yolov8m-pose.pt` in the COCO-17 convention. Each frame stores frame index, relative timestamp, keypoints with $[x, y, \text{confidence}]$, bounding box, score, and detection count. If multiple people are detected, the

largest bounding box is selected. The completed H5 is `data/pose2d_coco17_yolov8m_853_video_aligned.h5`.

5.2. 3D Pose

The 3D pose pipeline uses the cropped Kinect depth stream and the 2D COCO-17 pose detections. For each selected 2D keypoint, the pipeline searches the surrounding depth region, back-projects the selected depth pixel into camera coordinates, and records a 3D keypoint together with validity and quality-control statistics. The release H5 is:

`data/pose3d_coco17_depth_853_video_aligned.h5`.

Per-sample quality-control statistics are stored together with the generated 3D pose outputs.

5.3. Human Mesh and SMPL

The mesh file uses a frame-level ROMP/SMPL processing archive from cropped Kinect RGB videos. The compact H5 stores SMPL pose parameters, shape parameters, joints, projected 2D joints, camera translation, and validity indicators, rather than storing full vertices for every frame. This keeps the public mesh information compact while preserving the information needed to reconstruct meshes through the SMPL model [4]. The release H5 is:

`data/human_mesh_smpl_853_video_aligned.h5`.

5.4. DensePose

DensePose information is generated from the cropped RGB videos using the official Detectron2 DensePose implementation [5] with the chart-based IUUV model `densepose_rcnn_R_50_FPN_slx`, which predicts a body-part chart index I and continuous surface coordinates (U, V) for pixels inside the detected human region [1]. The release stores bounding boxes, scores, DensePose outputs, and UV fields. The release H5 is `data/densepose_rcnn_R50_FPN_853_video_aligned.h5`.

6. Action Annotations

6.1. Temporal Action Localization

The TAL annotation is stored in ActivityNet-style JSON at `annotations/tal_annotations_video_aligned/activitynet_tal_853.json`. Each sample contains relative-time segments, normalized segments, action indices, original action IDs, and metadata parsed from the sample name, including volunteer ID, scene name, scene ID, and sequence ID. The TAL release contains 853 samples, 8157 action segments, 34 unique labels, a duration range of 43–89 seconds, and 7–11 actions per sample. Absolute clock-time fields were removed.

Table 5. English Action QA examples from sample 0_Diningroom_1.

Type	Question	Answer
Existence	Did the participant sit down?	yes
Counting	How many times did the participant sit down?	1
Segment duration	How long did the first action, sitting down, last?	6 seconds
Temporal order	Does sitting down happen before walking toward the chair?	yes
Time-point action	What action was the participant doing at 3 seconds?	sitting down
Interval actions	What actions occur between 6 seconds and 26 seconds?	pouring water, drinking water, standing up
Longest segment	Which action segment lasted the longest in this sequence?	washing hands (10 seconds)

Table 6. Action QA type distribution.

Type	Count	Type	Count
existence	10399	count	9546
immediate order	9010	action at position	8157
action start time	8157	action end time	8157
segment duration	8157	total duration	7840
temporal location	7840	next action	7304
previous action	7304	temporal order	5118
timepoint action	2559	duration comparison	2557
actions between times	1706	first action	853
last action	853	longest segment	853
shortest segment	853	number of segments	853
action sequence	853		

6.2. Action Captioning

Captioning annotations are derived from the TAL segments and are provided in Chinese-only, English-only, and bilingual formats. Each sample contains dense segment captions and one sequence-level caption. The dense captions align one sentence with each timestamped action segment, while the sequence-level caption describes the ordered action routine. The captioning release contains 853 samples, 8157 action segments, and 34 action classes.

For example, one English sequence-level caption for sample 0_Diningroom_1 is: *The participant first sits down, then pours water, next drinks water, after that stands up, then walks toward the cabinet, then wipes the table, next throws away trash, after that washes hands, and finally walks toward the chair.*

Another English sequence-level caption, from sample 8_bedroom_13, is: *The participant gets out of bed, then stands up, then stretches, then walks to the window, then opens or closes the curtains, then opens or closes the window, then waters the plant, then walks to the bed, then sits down, and finally lies down.*

6.3. Action Question Answering

Action QA annotations are generated from the same relative-time TAL segments. XRFv2 Plus provides Chinese-only, English-only, and bilingual QA files. Each QA item includes

a question type, question, answer, answer type, optional time reference, source action indices, and evidence segments. The release contains 108,929 QA pairs across 21 question types.

Table 5 presents English QA examples, and Table 6 reports the Action QA type distribution.

The `total_duration` QA type is defined as the summed duration of all occurrences of the queried action within a sequence. It is not the duration of the whole input sequence.

7. Conclusion

XRFv2 Plus provides a video-aligned multimodal benchmark with standardized signal tensors, relative-time annotations, bilingual language annotations, and visual human information. The standardized 853-sample release is designed to support temporal action localization, action captioning, action question answering, multimodal fusion, video-feature learning, pose estimation, mesh reconstruction, DensePose prediction, and cross-modal human-sensing research. By explicitly documenting sampling rates, tensor conventions, device order, annotation derivation, and trimmed video cases, XRFv2 Plus makes XRFv2 easier to use and reproduce for both multimodal sensing and sensor-vision-language research.

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